Functions

Problem 1. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x)f(y) = f(x+y) + xy + x + y$$

for any $x, y \in \mathbb{R}$.

Problem 2. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y)) = f(x) + xy$$

for any $x, y \in \mathbb{R}$.

Problem 3. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m - n + f(n)) = f(m) + f(n)$$

for any $m, n \in \mathbb{Z}$.

Problem 4. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y) - f(x)) = f(xy) - xy$$

for any $x, y \in \mathbb{R}$.

Problem 5. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(x) + 2y) = 2x + 2f(f(y))$$

for any $x, y \in \mathbb{R}$.

Problem 6. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)f(y) + y^2) + f(x) = xy + f(y^2 + f(x))$$

for any $x, y \in \mathbb{R}$.

Problem 7. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m+n) + n \mid m + f(n)$$

for any $m, n \in \mathbb{N}$.

Problem 8. Find all functions $f : \mathbb{N} \to \mathbb{N}$ so that for every natural numbers m, n, $f(n) + 2mn + f(m)$ is a perfect square.